VOICE ARENA TAKE-HOME · Q1 · ONE-PAGE SUMMARY

DiarizBench: a diarization leaderboard for multilingual product teams

A benchmark design tested on three open systems and 24 recordings, and seven languages — with every number traceable to a saved run.

PURPOSE

A product team choosing a diarization engine is not asking which model has the lowest error rate. It is asking whether the system knows **who spoke when**, whether it keeps the **quiet participant** who says twelve seconds in a five-minute meeting, and whether it returns answers **fast enough** under the conditions the product actually runs in. The leaderboard turns those three questions into three published outputs instead of one score.

SETUP

Three open systems × 24 recordings × three corpora (Indic DiarBench 12, AMI-SDM 6, VoxConverse 6), seven languages, **205 minutes of scored audio** and 196 minutes of reference speaker-time. The contract was frozen before inference: zero collar, overlap scored, speaker count unknown, identical 16 kHz input, documented defaults, no tuning on pilot audio. **72/72 units completed with zero errors.**

PRINCIPAL FINDING — DER %, COLLAR SENSITIVITY, OVERLAP SCORED, UNKNOWN COUNT

MODE	SYSTEM	DER · 0 MS (95% CI)	DER · 250 MS Δ VS 0 MS	JER (95% CI)	EXACT COUNT	LOW-ACTIVITY RECALL	RTF
OFF	**CIIL Anuvani**	13.8 10.0 to 17.9	10.3 −3.5 pp	29.9 22.7 to 37.4	33% 8/24	47% 18/38	0.053 18.8×
OFF	**pyannote community-1**	20.4 15.5 to 25.3	16.7 −3.6 pp	38.6 30.3 to 47.2	38% 9/24	45% 17/38	0.072 13.9×
STR	**NVIDIA Nemotron-3 (streaming)**	21.6 15.7 to 27.6	18.8 −2.8 pp	32.5 23.4 to 41.6	58% 14/24	61% 23/38	0.079 12.7×

Collar: a window excluded around reference speech boundaries. Here **pyannote.metrics c = 0.25 s** removes 125 ms on each side; the second DER column is sensitivity, with its change in percentage points. Zero collar remains primary. Overlap and unknown speaker count are unchanged. Other scorers may define the collar differently.

> **Anuvani has the lowest observed DER: 13.8%.** Its offline gap against community-1 is +6.6 pp and survives Bonferroni correction and the 1 pp practical threshold. The +7.8 pp gap against streaming Nemotron is a comparison of configurations, not an offline-model ranking. The community-1/Nemotron DER difference is unresolved (95% CI −4.2 to +2.0); this does not establish equivalence. All averages are exploratory across native corpus conventions.

THREE DESIGN DECISIONS THAT CHANGED THE ANSWER

01

Score strictly, then publish the sensitivity

Zero collar is primary. A boundary collar reduces sensitivity to timestamp offsets, but can mask short turns. The saved 250 ms setting lowers DER by 2.8–3.6 pp; both views are shown, with the scorer's collar convention explicit.

02

Publish speaker coverage next to accuracy

Despite an unresolved DER gap, count accuracy (38% vs 58%) and low-activity recall (45% vs 61%) differ by 20.8 pp on count accuracy and 15.8 pp on low-activity recall, descriptively. Coverage superiority was not tested.

03

Never pool corpora silently

Native references and conversation content differ; the structural audit cannot separate their effects. Publish per-corpus results and exploratory aggregates. Keep offline and streaming ranks separate.

LIMITATION THAT MATTERS MOST

Exposure and sample size. Anuvani has Indic DiarBench corpus exposure; its parent also used AMI/VoxConverse. community-1 has AMI/VoxConverse exposure. Exact session overlap is unaudited. Two recordings per Indic language cannot support language-level rankings.

Mode and references. Nemotron ran streaming; the other two offline. No matched-mode rerun or listening-based annotation audit was completed. Timing is one-hardware, with different callable scopes; exact weight/env locks are incomplete. cpWER/tcpWER were not measured.

WITH MORE TIME

Add **cpWER and tcpWER** for speaker-attributed transcription, plus WER/CER, recognition-versus-attribution diagnostics, and overlap/VAD F1. These require human-checked speaker transcripts and timing, a fixed ASR backend, and a separate MeetEval protocol.

Expand to **DIHARD III, ICSI, CALLHOME, DISPLACE, AliMeeting, and CHiME-8**, subject to access and held-out split checks. Prioritize a matched offline Nemotron run, human reference QA, and controlled **separation, speaker similarity, clustering, and VAD/SAD** tracks (proposed, not measured).